Scalability & Future Plan

This document explains how the Electric Vehicle Market Analysis Using Tableau project can be enhanced to handle larger datasets, support additional users, improve dashboard performance, and introduce new analytical features. Future improvements may include updated EV market data, additional visualizations, advanced analytics, and stronger data security.

1. Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	The analysis depends on the available EV dataset and its existing fields.	New market developments or missing information may not be reflected in the dashboard.	High
2	The dashboards mainly provide descriptive analysis through charts and KPIs.	Users may have limited ability to forecast future sales or identify emerging trends.	Medium
3	Dashboard performance may decrease when larger datasets or more complex calculations are added.	Loading and filtering may become slower as the volume of data increases.	High

2. Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Designed for individual analysis and demonstration.	Publish the dashboard on Tableau Public for wider access, or use an appropriate shared Tableau platform if access controls are required.
2	Data Storage	Uses the currently available EV dataset.	Integrate larger datasets and maintain regularly updated, structured data sources.
3	Performance	Uses Tableau worksheets, filters, calculated fields, and dashboards.	Optimize calculations, remove unnecessary fields, and use extracts or suitable data connections where appropriate.
4	Security	Public sharing is possible through Tableau Public.	Avoid confidential data in public workbooks and use an access-controlled platform if private data is introduced.

3. Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Improve existing dashboards and verify all KPIs, filters, and visualizations.	Short term	Better readability and more reliable insights.
Phase 2	Integrate updated EV market data and additional manufacturer and country information.	1–2 months	More current and comprehensive market analysis.
Phase 3	Add sales forecasting and trend analysis using historical data.	3–4 months	Better understanding of potential future market trends.
Phase 4	Add advanced comparisons, sustainability indicators, and further dashboard optimization.	5–6 months	More detailed analysis and improved decision support.